

Product Innovation

Title: COVID-19 Detection System: AI-Powered Clinical Diagnostic Platform for COVID-19 Detection using Deep Learning and Explainable AI

1. Introduction

The presented product introduces an innovative AI-driven solution for the automated diagnosis of COVID-19 and related pulmonology conditions via chest X-ray images. Unlike other machine learning solutions that rely on one predictive algorithm, this system integrates several deep learning models, explainable AI components, and a number of scalable deployment options to ensure maximum interpretability, usability, and adaptability. In addition to delivering accurate predictions, the system's aim is to be reliable and useful to medical specialists when used in their clinical practices.

To train the proposed deep learning model, thousands of chest X-rays with labels were employed in this project. This data helps the model to identify several classes: COVID-19, normal lungs, viral pneumonia, and lung opacity. What makes this innovation different from traditional solutions is that this platform utilizes the power of a multi-model ensemble mechanism that involves the analysis of the same input image by several different models that reach a conclusion.

Along with improving the performance of the predictive process, this feature adds another layer of reliability to the diagnostic process since more than one model has to make the same decision for it to happen. Besides achieving the highest level of prediction accuracy, the system emphasizes the importance of being transparent and explainable. Clinicians expect a detailed explanation for each predicted class to help diagnose the patient and take further action.

A developed implementation of the current work is present at the following link:

<https://github.com/ssdafaef/PulmonaryAI>

2. Problem Statement

Current AI-based diagnostic solutions have faced some critical issues within the healthcare domain that limit their usage within clinical practice. One of the most prominent problems lies in using a single deep learning model that makes them prone to bias. Considering that any errors that the system makes will directly affect the health and life of patients, it is crucial to develop a reliable solution to deal with such a problem.

In addition, some diagnostic systems are primarily designed for research purposes and thus do not reflect real-life processes of working with medical data and making diagnoses. For example, they lack scalability and do not offer an opportunity to integrate the tool into existing systems and software solutions. In reality, there is a variety of steps required to be taken for diagnosing, such as data validation, collaboration, and structuring.

Another challenge of the current AI diagnostic solution is its inability to manage uncertainty. Current technologies tend to produce overconfident predictions that may influence practitioners'

decisions in a wrong way. Moreover, there is a lack of some practical features, like generating reports, classification of the disease severity level, and multi-language support.

Moreover, such systems do not use any approaches for uncertainty management or providing confidence scores, making predictions highly overconfident. Another important point is the lack of other useful features like automated report generation, severity categorization, or multilinguality, which makes these solutions not usable in practice. Hence, the aim of this project is to develop a system that will have all those shortcomings addressed.

3. Proposed Solution

The proposed system represents a hybrid AI solution incorporating the latest advancements in deep learning, explainable artificial intelligence, and deployment frameworks in order to deliver a full-fledged clinical decision support system for chest X-ray image assessment. As the core component of the system, a convolutional neural network with a special attention module is used. The chosen network structure is ResNet50.

For increased reliability of the solution, a multi-model ensemble method will be utilized. It will consist of several independent neural networks performing predictions based on a provided image. Then, using a certain algorithm, a consensus between the predictions generated by individual networks will be achieved. Thus, only the most confident predictions from different networks will make their way to the final result. This is inspired by the process of collaborative work of physicians in reality, where a team of specialists contributes to a final decision.

The problem of interpretability and explanation of predictions will also be solved using explainable artificial intelligence techniques. Grad-CAM algorithms will allow generating visual explanations represented by the heatmaps showing what parts of the image influenced the decision of the classifier. Moreover, a separate language model will be able to generate a natural-language explanation of a predicted diagnosis. It will be accompanied by the reasoning behind the prediction, its consequences, and possible further actions.

Automated generation of reports with all predictions and explanations will be implemented as another crucial feature. Reports generated by the solution will be in the PDF format, containing all data regarding the analysis of a chest X-ray image, including severity level. Moreover, there will be multilingual capabilities of the solution, along with a severity classification system.

4. System Workflow

The operation of the system involves a pipeline workflow simulation that resembles a realistic diagnostic scenario. The procedure starts by uploading a chest X-ray image through the web application. It is designed to allow easy interaction for both experienced and non-experienced users.

Once the image is uploaded, it is preprocessed in the backend environment. It includes image size rescaling, normalization of the image's pixel values, and conversion to a specific format. Proper preprocessing is vital for achieving consistent and reliable performance of AI models.

Further, the image is subjected to the analysis of several AI models. Each of them independently provides probability outputs related to different classes. Outputs of different models are combined using the technique of voting. If there is consistent behavior between the models, a certain classification result with high confidence level is obtained. However, in case of discrepancy, the

system informs about the inconsistency and suggests additional analysis. After the prediction, the system uses explainable AI and provides heatmaps highlighting the most influential areas in the image. Further, an AI-driven natural language explanation of the results is generated by the language model.

Additionally, the system performs severity classification for COVID-19 infection cases. As a final step, all information is aggregated and provided via the interface. A PDF document containing information related to the diagnosis is prepared automatically and ready for download.

5. Key Innovations

- AttentionResNet50 with a custom attention mechanism for COVID-19 classification
- Hybrid inference approach using NVIDIA Triton Inference Server and local Keras models
- Explainable AI based on the Grad-CAM visualization of every prediction
- Voting between multiple inference branches based on the confidence in prediction
- Ai-powered explanation using a language model (LLM; Google Gemini)
- Automated PDF report generation with multi-language support
- Four-class classification of Chest X-rays: COVID-19, Normal, Viral Pneumonia, Opacity
- Accuracy ~94%, ~0.97 AUC-ROC on 21,165 chest X-ray images
- Severity classification for COVID-19 infections (mild, moderate, severe)
- Fully deployed system using Docker Compose
- Model serving on NVIDIA Triton Inference Server
- End-to-end AI pipeline for uploading images and generating PDF reports

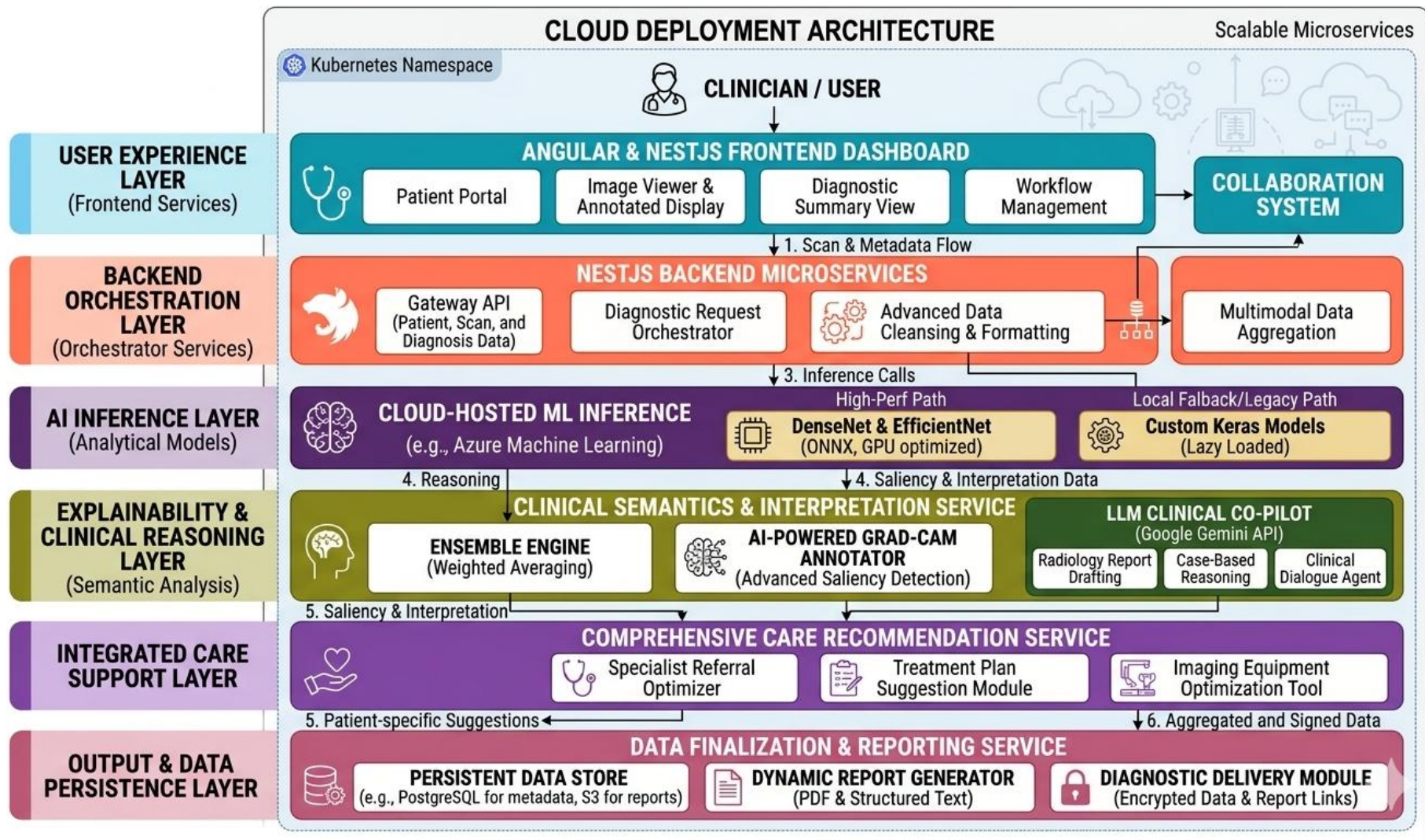


Figure 61: Architecture Diagram of model

Medical AI Diagnostic Portal

Agentic AI integration for hospital resource optimization and triage support.

Analyses
3

Status
Ready

Size
0.04 MB



Change Image

Run AI Prediction

Copy Summary

Clear Session

Diagnosis Results

High confidence

Classification:

COVID

AI Confidence:

91.98%

What To Do Next

Likely Condition: COVID-19 pattern suspected

Urgency: Prompt physician review recommended within the same day.

Recommended Actions

- Isolate patient as per local infection prevention protocol.
- Check oxygen saturation, respiratory rate, and temperature at triage.
- Correlate with symptoms and confirm using RT-PCR/rapid antigen as indicated.

Red Flags

- SpO2 below 94%
- Increasing shortness of breath
- Persistent high fever or chest pain

Follow-Up: Reassess clinically in 12 to 24 hours or earlier if symptoms worsen.

Figure 62: X-ray prediction output showcasing result

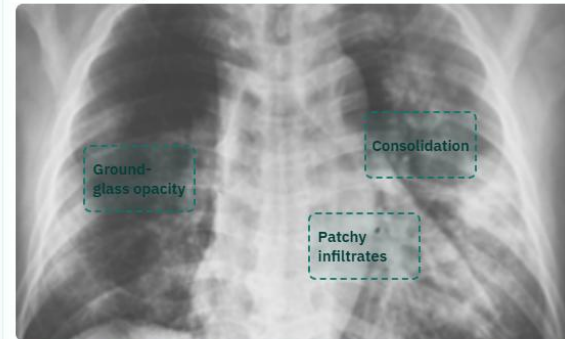
Medical AI Diagnostic Portal	123 Trichy, India
Doctor Name	Specialization
Dr. Olivia Greene	Doctor
Visit Date	Patient Name
05 / 03 / 2026	Sarah Anderson
Birth Date	Medical Number
01 / 01 / 1989	MA567891
IHI	Phone
5556-9669-9654-7788	+1 (555) 789-0123
Email	Hospital Contact
s.anderson@mail.com	+91 (XXX) 123-4567
Hospital Email	Hospital Site
info@Medical_AI_Diagnostic.com	www.Medical_AI_Diagnostic.com

Source: remote-llm | Latency: 9296 ms | Gemini-backed LLM

Impression: Multifocal, predominantly peripheral ground-glass opacities and consolidations are present bilaterally, significantly more pronounced in the left lung. These findings are highly suggestive of viral pneumonia, such as COVID-19, given the clinical context and high AI confidence. Correlation with RT-PCR testing and clinical symptoms is strongly recommended.

Narrative Summary: The frontal chest radiograph demonstrates extensive, patchy opacities distributed throughout both lung fields. There is a notable peripheral and mid-to-lower zone predominance, which is characteristic of atypical or viral pneumonias. The left lung shows more confluent areas of consolidation compared to the right, particularly in the mid-lung zone. No significant pleural effusions or pneumothorax are identified. The cardiomediastinal silhouette remains within normal limits for size and configuration, and the visualized osseous structures are unremarkable.

LLM Pattern Map



- **Ground-glass opacity:** Patchy peripheral opacities in the right mid-to-lower lung zone.

Figure 63: The LLM model showcasing overlay of detected area

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Gordon BETA

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit BETA

Docker Hub

Docker Scout

Extensions

Upgrade plan

seaaai

D:\seaaai

View configurations

api	INFO:	127.0.0.1:40050	- "GET /health HTTP/1.1" 200 OK	
afeefahmed/seaaai:8010:8010	api-client	INFO:	127.0.0.1:52096 - "GET /health HTTP/1.1" 200 OK	
		INFO:	127.0.0.1:53560 - "GET /health HTTP/1.1" 200 OK	
triton-server	api	INFO:	127.0.0.1:40076 - "GET /health HTTP/1.1" 200 OK	
seaaai-triton-server:8000:8000	api-client	INFO:	127.0.0.1:56450 - "GET /health HTTP/1.1" 200 OK	
Show all ports (3)		api	INFO:	127.0.0.1:50136 - "GET /health HTTP/1.1" 200 OK
api-client	api-client	INFO:	127.0.0.1:49298 - "GET /health HTTP/1.1" 200 OK	
seaaai-api-client:8011:8001		api	INFO:	127.0.0.1:53052 - "GET /health HTTP/1.1" 200 OK
frontend	api-client	INFO:	127.0.0.1:44514 - "GET /health HTTP/1.1" 200 OK	
seaaai-frontend:3000:80		INFO:	127.0.0.1:43372 - "GET /health HTTP/1.1" 200 OK	
	api	INFO:	127.0.0.1:43338 - "GET /health HTTP/1.1" 200 OK	
	api-client	INFO:	127.0.0.1:36108 - "GET /health HTTP/1.1" 200 OK	
	api	INFO:	127.0.0.1:49668 - "GET /health HTTP/1.1" 200 OK	
	api-client	INFO:	127.0.0.1:37116 - "GET /health HTTP/1.1" 200 OK	
	api	INFO:	127.0.0.1:34674 - "GET /health HTTP/1.1" 200 OK	
	api-client	INFO:	127.0.0.1:59560 - "GET /health HTTP/1.1" 200 OK	
		INFO:	127.0.0.1:60654 - "GET /health HTTP/1.1" 200 OK	
	api	INFO:	127.0.0.1:48212 - "GET /health HTTP/1.1" 200 OK	
	api-client	INFO:	127.0.0.1:40018 - "GET /health HTTP/1.1" 200 OK	
	api	INFO:	127.0.0.1:59322 - "GET /health HTTP/1.1" 200 OK	

Engine running

RAM 3.46 GB CPU 0.37% Disk: 66.77 GB used (limit 1006.85 GB)

Terminal v4.71.0

Figure 65: All the container inside docker container

1. Dataset and Training Details

The creation of the proposed COVID-19 Detection System is possible because of the [COVID-19 Radiography Database](#), which is the commonly-used dataset for medical image classification tasks. It includes numerous chest radiography images grouped into several classes: COVID-19, Normal, Viral Pneumonia, and Lung Opacity. The presence of over 21,000 images makes it diverse enough to train the system with deep learning methods.

The dataset has training, validation, and testing samples to ensure the possibility of reliable evaluation of the system's performance. For example, the training subset is utilized to train models and learn necessary patterns, while the validation one monitors the learning process and prevents overfitting. At last, test data is used to assess the results and estimate the performance in real-world conditions.

To enhance the performance of models, multiple preprocessing techniques are applied. They include resizing images into standard input dimensions (224×224 pixels) and pixel values normalization. Some other techniques used are rotations, flips, and scaling of images to diversify data. Finally, because of class imbalance, class weighting was applied during the learning stage to ensure fair results.

Deep learning models are trained using TensorFlow library in Google Colab equipped with GPU. Transfer learning is used to pretrain the models on pretrained architectures such as ResNet50 and DenseNet121. The best performance of models trained with transfer learning and without it are compared. Moreover, class weighting and standard training techniques are also explored. The accuracy score reaches about 94% when using the most productive approach. Thus, the selected combination of techniques yields impressive results and ensures an accurate system.

In general, with an extensive dataset, numerous preprocessing techniques, transfer learning, and GPU utilization, the developed system is highly accurate and reliable.

2. Deployment and Scalability

To provide scalability and portability to the developed system, a number of techniques can be used. They include the creation of self-contained units, packaging of components into containers, and implementing cloud computing capabilities. All mentioned elements are present in the design of the system, making it scalable and portable.

A high-performance inference server is utilized to efficiently process numerous requests and perform in real-time environment. Additionally, because of the hybrid architecture, the system works in environments with or without GPUs. The use of container orchestration makes it easy to scale the system up to handle higher loads.

By implementing such technologies as container orchestration, the system can easily scale to process higher volumes of data. Therefore, the solution can be deployed in hospitals, scientific centers, and even in remote medical centers. Generally speaking, the deployment approach enables the creation of an AI tool that is powerful and pragmatic.

3. Real-World Impact

With its help, the diagnosis of such diseases as coronavirus infection, lung inflammation, and other related conditions can be carried out with precision and accuracy. Furthermore, the AI-driven diagnostic tool will make it easier for specialists to make informed choices since the system offers both the results and explanations. It may serve as an additional aid to those parts of the world where experienced radiologists cannot be found.

The use of such a system can positively affect healthcare through improved speed and quality of diagnostics. Moreover, the possibility of generating structured reports and evaluating the severity of symptoms increases the efficiency of the entire diagnostic process.

4. Conclusion

This paper has offered an insight into developing a powerful system designed to help people get reliable diagnoses. Such a tool combines various modern approaches used in machine learning and can provide useful solutions for medical diagnostics.

The project can be seen as an example of the successful application of artificial intelligence in solving complex problems and creating helpful applications for everyday practice.

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Demo Video as per Guidelines

The screenshot displays a web application for COVID-19 diagnosis. On the left, a video feed shows a user. The main area features a chest X-ray image with a 'Change Image' button below it. A 'Run AI Prediction' button is prominently displayed. Below this are 'Copy Summary' and 'Clear Session' buttons. The 'Report Profile' section contains fields for Hospital Name (Medical AI Diagnostic Portal), Hospital Address (123 Trichy, India), Doctor Name (Dr. Olivia Greene), Specialization (Doctor), Visit Date, and Patient Name. The 'Diagnosis Results' section shows a 'Medium confidence' level, a 'COVID' classification, and an 'AI Confidence' of 70.51%. It also includes a 'What To Do Next' section with 'Likely Condition: COVID-19 pattern suspected', 'Urgency: Prompt physician review recommended within the same day.', 'Recommended Actions' (Isolate patient, Check oxygen saturation, Correlate with symptoms), and 'Red Flags' (SpO2 below 94%, Increasing shortness of breath, Persistent high fever or chest pain). A 'Follow-Up' instruction is also present. At the bottom, there are 'AI Clinical Insight' and 'Generate Insight' buttons, along with 'TXT' and 'PDF' download buttons. A disclaimer at the bottom states: 'Research use only. Not for final clinical diagnostic use.'

Fig 66 : Vedio Drive Link : <https://drive.google.com/drive/folders/1HYollCRsFegY67pfavab6J0K13X4rnKf?usp=sharing>

Github Repo and Fork

The screenshot displays the GitHub repository page for 'PulmonaryAI' by user 'ssdafeef'. The repository is public and has 22 forks and 0 stars. The main branch is 'main'. The repository contains three files: 'seaaai', 'LICENSE', and 'README.md'. The 'README.md' file is selected, showing the title 'COVID-19 Detection System' and a description: 'An AI-powered clinical diagnostic platform for COVID-19 detection using chest X-ray analysis.' A disclaimer is also present: 'Disclaimer: This COVID-19 Detection System is intended for educational and research use only. It is not a certified medical device and must not be used as a substitute for professional clinical diagnosis.' The repository has 8 commits and 1 contributor. The right sidebar shows the 'About' section with no description, website, or topics provided, and the 'Releases' section with no releases published.

ssdafeef / PulmonaryAI

Type / to search

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

PulmonaryAI Public

Pin Watch 0 Fork 22 Star 0

main 1 Branch 0 Tags

Go to file T Add file <> Code

ssdafeef Add README for COVID-19 Detection System 1d0b380 · yesterday 8 Commits

File	Commit Message	Commit Date
seaaai	Revise README for PneumoAI project overviewreadme update	yesterday
LICENSE	Initial commit	yesterday
README.md	Add README for COVID-19 Detection System	yesterday

README MIT license

COVID-19 Detection System

An AI-powered clinical diagnostic platform for COVID-19 detection using chest X-ray analysis.

Disclaimer: This COVID-19 Detection System is intended for educational and research use only. It is not a certified medical device and must not be used as a substitute for professional clinical diagnosis.

Overview

About

No description, website, or topics provided.

- Readme
- MIT license
- Activity
- 0 stars
- 0 watching
- 22 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 1

Fig 67: GitHub Repository Overview (PneumoAI Project) — <https://github.com/ssdafeef/PulmonaryAI>

Review 1: Quality of the Module Developed

First of all, an essential aspect of the developed module is the presence of a structure in the system and the successful use of several advanced technologies within the model. It should be mentioned that the process of creating the model was performed on the basis of a rather rich COVID-19 Radiography Database, which includes many samples of chest radiograph images. The interaction with such a resource allowed considering important aspects, such as class distributions, data preparation, and training processes, contributing greatly to the creation of the proposed system.

Another crucial feature of the module is high computational efficiency due to the possibility to work with GPU. Training models in Google Colab using GPU resources allowed performing faster tests and avoiding problems related to the necessity to process big amounts of data and train complicated structures of deep learning models.

Moreover, it is important to note that another aspect of the quality of the module concerns its architecture. In the context of the described project, one can observe how the system successfully combines such important components as data preprocessing, model prediction, explaining AI, and report generation. Moreover, each component performs its function while improving the performance of the overall system.

In addition to the previous aspects, it is vital to note the presence of the feature of the generated explanations by the system. The ability to see visual explanations of results and use the information provided by AI about the disease increases the quality of the tool significantly.

Overall, it should be noted that the created project demonstrates the high quality of its modules since theoretical ideas are combined with the implementation of the solution..

COVID-19 Radiography Database

Data Card Code (511) Discussion (27) Suggestions (0)

*COVID-19 CHEST X-RAY DATABASE

A team of researchers from Qatar University, Doha, Qatar, and the University of Dhaka, Bangladesh along with their collaborators from Pakistan and Malaysia in collaboration with medical doctors have created a database of chest X-ray images for COVID-19 positive cases along with Normal and Viral Pneumonia images. This COVID-19, normal and other lung infection dataset is released in stages. In the first release we have released 219 COVID-19, 1341 normal and 1345 viral pneumonia chest X-ray (CXR) images. In the first update, we have increased the COVID-19 class to 1200 CXR images. In the 2nd update, we have increased the database to 3616 COVID-19 positive cases along with 10,192 Normal, 6012 Lung Opacity (Non-COVID lung infection) and 1345 Viral Pneumonia images. We will continue to update this database as soon as we have new x-ray images for COVID-19 pneumonia patients.

**COVID-19 data:

COVID data are collected from different publicly accessible dataset, online sources and published papers.
 -2473 CXR images are collected from padchest dataset[1].
 -183 CXR images from a Germany medical school[2].

COVID 2 directories	Lung_Opacity 2 directories	Normal 2 directories	Viral Pneumonia 2 directories	README.md.txt 3.35 kB
COVID.metadata.xlsx	Lung_Opacity.metadat...	Normal.metadata.xlsx	Viral Pneumonia.meta...	

- COVID-1013.png
- COVID-1016.png
- COVID-1017.png
- COVID-1018.png
- COVID-1019.png
- COVID-102.png
- COVID-1020.png
- COVID-1021.png
- COVID-1022.png
- COVID-1023.png
- COVID-1024.png
- ... 3586 more
- Lung_Opacity
- Normal
- Viral Pneumonia
- README.md.txt
- COVID.metadata.xlsx
- Lung_Opacity.metadat
- Normal.metadata.xlsx
- Viral Pneumonia.metac

Summary

- 42.3k files
- 16 columns

Figure 68: All the different dataset images

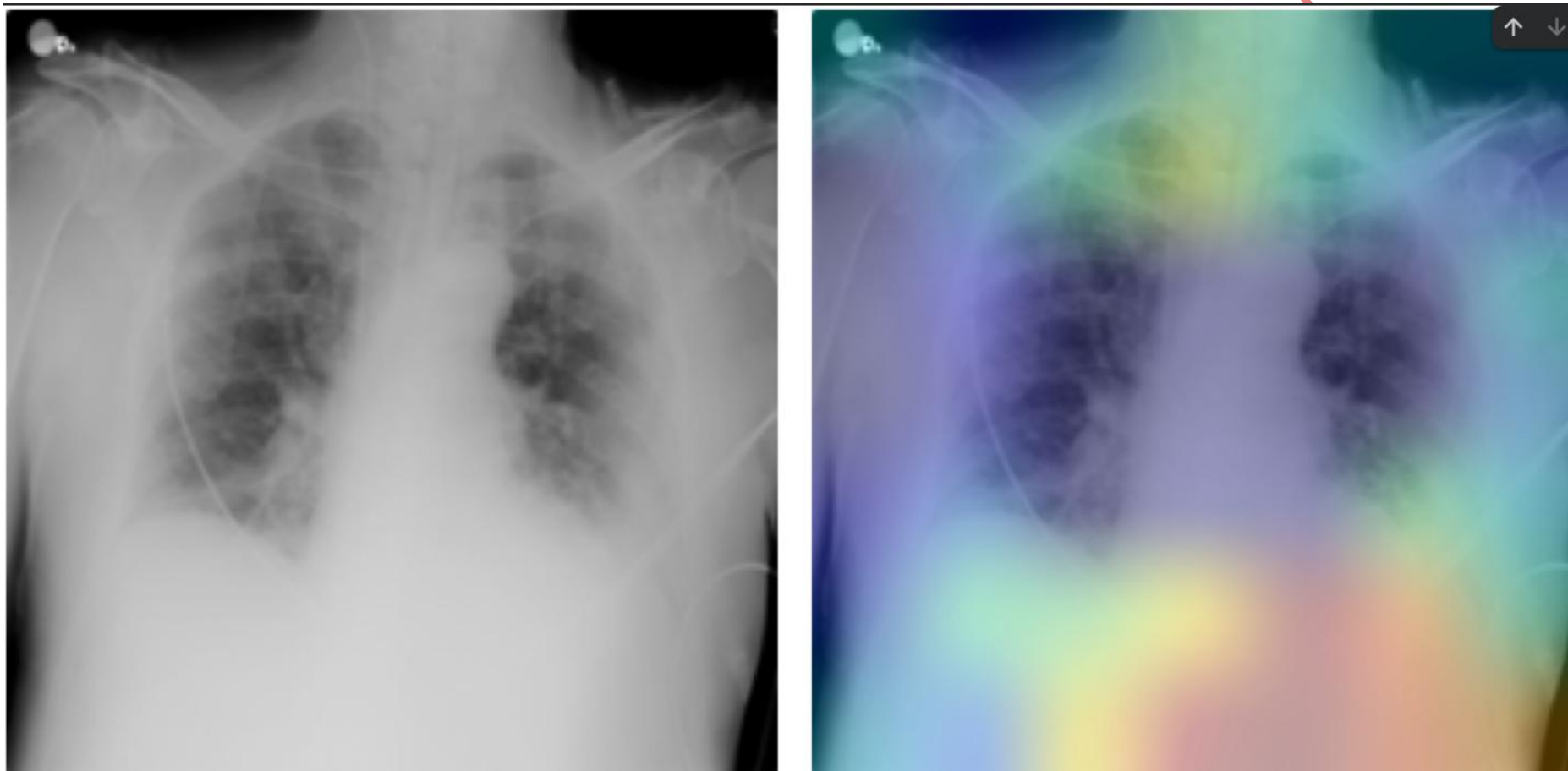


Figure 69: Untrained model false detection before fine tuning

AFFEEF AHMED

Step 1. Original X-ray



Step 2. Grad-CAM novelty



Figure 70.: Huge detection improvement after fine-tuning the model

AFFEEF AHMED

Review 2: Containerization and Deployment

One of the most important features of the developed COVID-19 Detection System is the presence of clear mechanisms for containerization and deployment which make the system easily reproducible, portable and fit for productive use.

During the process of system creation, a special focus was put on following modern software architecture and deployment trends. Therefore, containerization was used in accordance with the modern principles based on Docker and Docker Compose technologies.

System Architecture:

The developed COVID-19 Detection System can be viewed as a multi-component architecture including the following parts:

- **Frontend:** React app that uses Vite for development and optimized building
- **Backend:** FastAPI server for images preprocessing, orchestration and API management
- **Model Inference:** NVIDIA Triton Inference Server for efficient, high-performance inference of models
- **Local Model:** As a fallback, Keras model can be used to ensure more reliable results

All of these components were packaged separately into containers by means of Docker technology. It guarantees that no external dependencies exist, and all necessary libraries are packed.

Docker Compose Orchestration

In order to facilitate the launch of multiple services, Docker Compose technology was used. The file docker-compose.yml describes services, their configuration, dependencies and networking. The system can be started with the following command:

```
docker compose up -d
```

This command does the following:

- Building and launching of Triton Inference Server with COVID classifier model loaded
- Starting a FastAPI server with correct health check setup and dependencies
- Launching React application
- Setting up inter-service communication and volumes for model loading

NVIDIA Triton Model Serving

When deploying a system, production-ready NVIDIA Triton Inference Server is used for efficient inference of pre-trained models. For setting up the service, the corresponding Docker image from NVIDIA repository is downloaded first:

```
docker pull nvcr.io/nvidia/tritonserver:23.10-py3
```

Service Endpoints

After deployment, all service endpoints can be accessed using the following URLs:

- **Frontend:** <http://localhost:3000>— React Vite application for uploading X-rays and visualizing results
- **Backend API:** <http://localhost:8010/docs> — FastAPI documentation with prediction endpoint
- **Triton Server:** <http://localhost:8001>— NVIDIA Triton Inference Server for models inference (HTTP), <http://localhost:8002> (gRPC)

Benefits of Deployment

The following are the key benefits of this deployment solution:

- **Consistency:** Standardized execution in development, testing, and production stages
- **Mobility:** All services can be executed on any device that supports Docker and Docker Compose
- **Scalability:** Individual scaling of services based on demand (front-end, back-end, inference)
- **Resilience:** Health checks guarantee the availability of services; fault tolerance is provided by the mixed inference pipeline using Triton and local Keras
- **Deployment Ready:** Enterprise-level orchestration that is capable of being deployed to cloud-based systems or on-site infrastructure

Medical AI Diagnostic Portal

Agentic AI integration for hospital resource optimization and triage support.

Analyses
3

Status
Ready

Size
0.04 MB



Change Image

Run AI Prediction

Copy Summary

Clear Session

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High confidence

Classification:

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AI Confidence:

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What To Do Next

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Recommended Actions

- Isolate patient as per local infection prevention protocol.
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Red Flags

- SpO2 below 94%
- Increasing shortness of breath
- Persistent high fever or chest pain

Follow-Up: Reassess clinically in 12 to 24 hours or earlier if symptoms worsen.

Figure 71: Main Dashboard

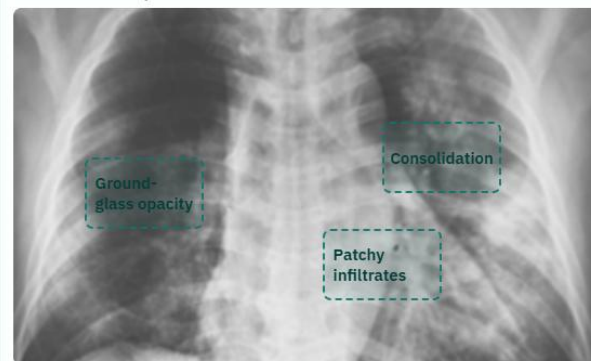
Medical AI Diagnostic Portal	123 Trichy, India
Doctor Name	Specialization
Dr. Olivia Greene	Doctor
Visit Date	Patient Name
05/03/2026	Sarah Anderson
Birth Date	Medical Number
01/01/1989	MA567891
IHI	Phone
5556-9669-9654-7788	+1 (555) 789-0123
Email	Hospital Contact
s.anderson@mail.com	+91 (XXX) 123-4567
Hospital Email	Hospital Site
info@Medical_AI_Diagnostic.com	www.Medical_AI_Diagnostic.com

Source: remote-llm | Latency: 9296 ms | Gemini-backed LLM

Impression: Multifocal, predominantly peripheral ground-glass opacities and consolidations are present bilaterally, significantly more pronounced in the left lung. These findings are highly suggestive of viral pneumonia, such as COVID-19, given the clinical context and high AI confidence. Correlation with RT-PCR testing and clinical symptoms is strongly recommended.

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LLM Pattern Map



- **Ground-glass opacity:** Patchy peripheral opacities in the right mid-to-lower lung zone.

Figure 72: LLM model Insights

LLM Annotated Image

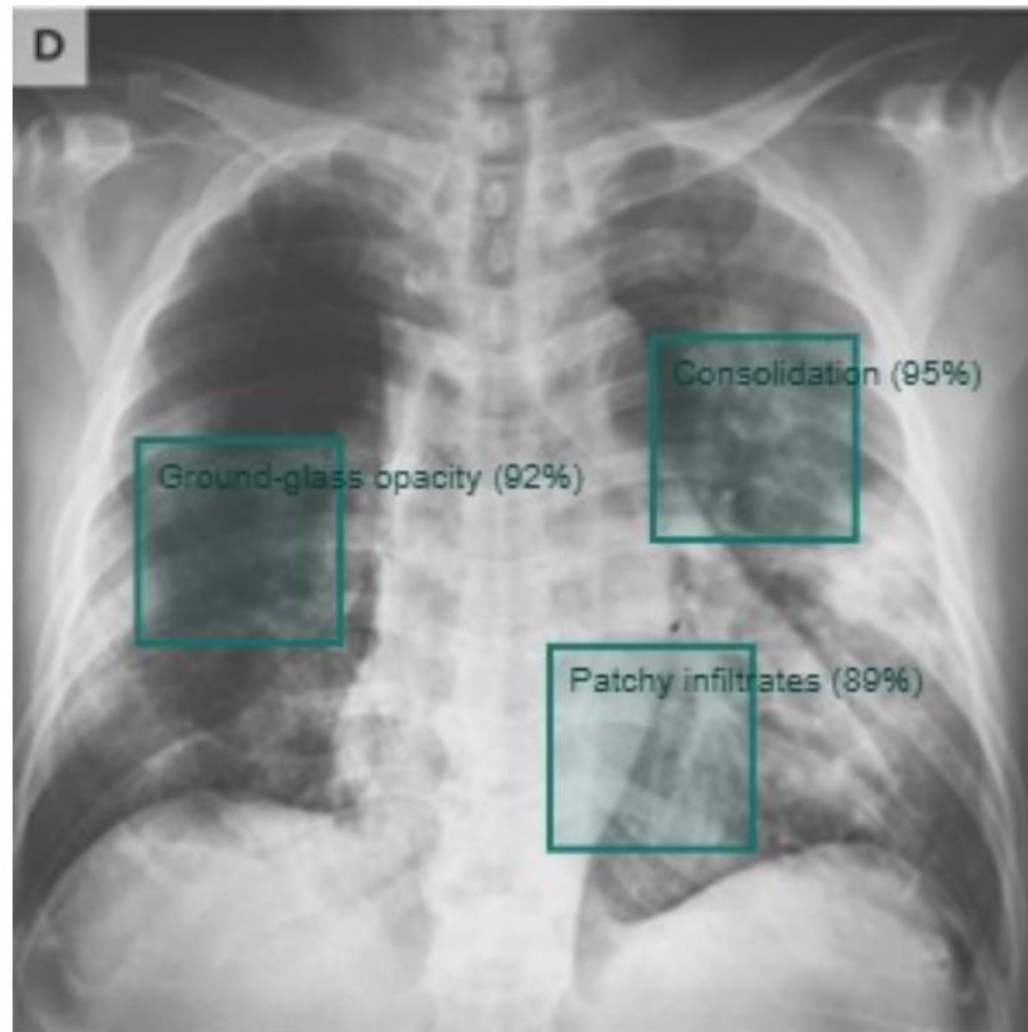


Figure 73: Model Overlay of Detected virus

docker.desktop

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BETA

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit

BETA

Docker Hub

Docker Scout

Extensions

seaaai

D:\seaaai

View configurations

api afeefahmed/seaaai 8010:8010	api INFO: 127.0.0.1:40050 - "GET /health HTTP/1.1" 200 OK
triton-server seaaai-triton-server 8000:8000 Show all ports (3)	api INFO: 127.0.0.1:40076 - "GET /health HTTP/1.1" 200 OK
api-client seaaai-api-client 8011:8001	api-client INFO: 127.0.0.1:52096 - "GET /health HTTP/1.1" 200 OK
frontend seaaai-frontend 3000:80	api-client INFO: 127.0.0.1:53560 - "GET /health HTTP/1.1" 200 OK
	api INFO: 127.0.0.1:50136 - "GET /health HTTP/1.1" 200 OK
	api-client INFO: 127.0.0.1:49298 - "GET /health HTTP/1.1" 200 OK
	api INFO: 127.0.0.1:53052 - "GET /health HTTP/1.1" 200 OK
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	api-client INFO: 127.0.0.1:43372 - "GET /health HTTP/1.1" 200 OK
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	api-client INFO: 127.0.0.1:60654 - "GET /health HTTP/1.1" 200 OK
	api INFO: 127.0.0.1:48212 - "GET /health HTTP/1.1" 200 OK
	api-client INFO: 127.0.0.1:40018 - "GET /health HTTP/1.1" 200 OK
	api INFO: 127.0.0.1:59322 - "GET /health HTTP/1.1" 200 OK

Engine running

RAM 3.46 GB CPU 0.37% Disk: 66.77 GB used (limit 1006.85 GB)

Terminal

v4.71.0

Figure 74: All containers running in docker